

Probability
Chapter 2 - Module 5

Fundamental Limit Theorems

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There are many tools that we need in the study of probability. Included among these are a number of theoretical results that will be very useful from time to time. We begin with two easy theorems.

Prop 2.6 (Markov's Inequality) Let X be a non-negative random variable. Then for any $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Proof. We will prove the result in the case the X is a continuous random variable.

In this case

$$\begin{aligned} E[X] &= \int_0^{\infty} x f(x) dx \\ &= \int_0^a x f(x) dx + \int_a^{\infty} x f(x) dx \\ &\geq \int_a^{\infty} x f(x) dx \\ &\geq \int_a^{\infty} a f(x) dx \\ &= a \int_a^{\infty} f(x) dx \\ &= a P(X \geq a). \end{aligned}$$

So

$$P(X \geq a) \leq \frac{E[X]}{a}$$

for any positive number a .

From this we can immediately prove the following.

Prop. 2.7 (Chebyshev's Inequality) Let X be a random variable with mean μ and variance σ^2 .

Then

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Proof. Since $(X - \mu)^2$ is non-negative, we may apply Markov's inequality to show that

$$P((X - \mu)^2 \geq k^2) \leq \frac{\sigma^2}{k^2}$$

and so

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Example 1. Suppose that the Ajax company produces a widget that weighs, on average, 10lbs.

Question: What can we say about the probability that a widget will weight at least 15 lbs?

Answer: Let X denote the weight of a random widget and let be a random variable with $E[X] = 10$. Now in all cases $X \geq 0$.

Then by Markov's inequality

$$P(X \geq 15) \leq \frac{E[X]}{15} = \frac{10}{15} = \frac{2}{3} = .667.$$

Question: Can we improve on this if we know that $\text{Var}(X) = 4$?

Answer: We will use Chebychev's inequality. Chebychev's inequality tells us that

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

So

$$P(X \geq 15) = P(X - 10 \geq 5) \leq P(|X - 10| \geq 5) \leq \frac{\sigma^2}{k^2} = \frac{4}{25} = .16.$$

Theorem. (Strong Law of Large Numbers) Let $X_1, X_2, \dots$ be an infinite sequence of iid random variables. Assume that $E[|X|] < \infty$ and that $\mu = E[X]$.

Then

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mu \text{ as } n \rightarrow \infty$$

with probability one.

So what does this mean?

To fully answer that question we would need to do 2 things.

1. Create the set (space) $H = \{(x_1, x_2, x_3, \dots)\}$ of all infinite sequences of real numbers.
2. Create a probability function over this space based on the probability function for X_1 .

With these two items in hand we could then make the following statement.

$$P\left((x_1, x_2, \dots) \mid \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n x_i = \mu\right) = 1.$$

This is as much detail as I will provide in this course, but the following is a good reference.

Reference. Probability and Random Processes by Geoffrey Grimmett and David Stirzaker, third edition, Oxford University Press, 2001.

Let's look at an example in Mathematica.

We will motivate this with some simple calculations using normally distributed random variables.

To begin, let $Z_1, Z_2, \dots, Z_n$ be iid standard normal random variables. Then by Module 4

$$\frac{1}{n} \sum_{i=1}^n Z_i \sim N \left(0, \sigma^2 = \frac{1}{n} \right).$$

Then by the Strong Law of Large Numbers

$$\frac{1}{n} \sum_{i=1}^n Z_i \rightarrow 0$$

and the variance goes to zero as well.

On the other hand,

$$\sum_{i=1}^n Z_i \sim N (0, \sigma^2 = n)$$

and while the mean is always zero, the variance increases without bound.

Question: How should we rescale the sum so that

$$\lim_{n \rightarrow \infty} c_n \sum_{i=1}^n Z_i$$

converges to a fixed non-degenerate distribution?

Answer:

$$\text{Var} \left(c_n \sum_{i=1}^n Z_i \right) = c_n^2 \sum_{i=1}^n \text{Var} (Z_i) = c_n^2 \sum_{i=1}^n 1 = nc_n^2,$$

So if we let $c_n = \frac{1}{\sqrt{n}}$, then

$$\text{Var} \left(c_n \sum_{i=1}^n Z_i \right) = \text{Var} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \right) = \frac{n}{n} = 1.$$

Question: How should this be adjusted if $X_1, X_2, \dots, X_n$ are iid normal random variables with mean μ and variance σ^2 .

Answer: Since $\frac{X_i - \mu}{\sigma} \sim N(0, 1)$ then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma} \sim N(0, 1)$$

as well.

The amazing fact is that if we consider limit of distributions, then this generalizes to random variables $X_1, X_2, \dots, X_n$ that are not normally distributed.

Theorem. (Central Limit Theorem) Let $X_1, X_2, \dots$ be iid random variables with mean μ and variance σ^2 . Then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma} \rightarrow N(0, 1)$$

in distribution, which means that for any real number a that

$$P \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma} \leq a \right) \rightarrow \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-x^2} dx.$$

An amazing observation is that this holds even when X_i is a discrete random variable.

Example 2. Suppose that X_i is a Bernoullie random variable with probability of success $p = .6$ for $n = 1, 2, \dots, 40$. Estimate the probability of 27 successes.

We recall that X_i is 1 with probability p and 0 with probability $1 - p$. So let

$$Y = \sum_{i=1}^n X_i.$$

The question then is what is $P(Y = 27)$.

Now Y is a binomial random variable and so the direct answer in the general case is that

$$P(Y = k) = p(k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Let's also use the Central Limit Theorem to estimate this probability.

We need the following.

1. $E[Y] = np$.
2. $\text{Var}(Y) = np(1 - p)$

So

$$\begin{aligned} P(Y = k) &= P(k - .5 \leq Y \leq k + .5) \\ &= P\left(\frac{k - .5 - np}{\sqrt{\text{Var}(Y)}} \leq \frac{Y - np}{\sqrt{\text{Var}(Y)}} \leq \frac{k - np + .5}{\sqrt{\text{Var}(Y)}}\right) \\ &\approx \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx, \end{aligned}$$

where

$$a = \frac{k - .5 - np}{\sqrt{\text{Var}(Y)}} \quad \text{and} \quad b = \frac{k + .5 - np}{\sqrt{\text{Var}(Y)}}.$$

Let's look at a Mathematica Notebook.

Question: Can we use this to estimate the probability that $Y = 27.2$?

Answer: No, but why not?

Let X be a binomial random variable with parameters n and p . And let $\lambda = np$. Then

$$\begin{aligned}
 P(X = i) &= \binom{n}{i} p^i (1-p)^{n-i} \\
 &= \frac{n!}{i! (n-i)!} p^i (1-p)^{n-i} \\
 &= \frac{n!}{i! (n-i)!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i} \\
 &= \frac{n(n-1) \cdots (n-i+1)}{n^i} \frac{\lambda^i}{i!} \frac{(1 - \lambda/n)^n}{(1 - \lambda/n)^i}.
 \end{aligned}$$

So

$$P(X = i) = \frac{n(n-1) \cdots (n-i+1)}{n^i} \frac{\lambda^i (1 - \lambda/n)^n}{i! (1 - \lambda/n)^i}.$$

Now when n is large and p is small then $\lambda/n = p$ is small. So

$$(1 - \lambda/n)^n \approx e^{-\lambda}$$

$$(1 - \lambda/n)^i \approx 1$$

and

$$\frac{n(n-1) \cdots (n-i+1)}{n^i} \approx 1.$$

So

$$P(X = i) \approx e^{-\lambda} \frac{\lambda^i}{i!}.$$

Recall that if $P(Y = i) = e^{-\lambda} \frac{\lambda^i}{i!}$, then Y is called a Poisson random variable with parameter λ .